AttriSense: An Explainable, Fair, and Resilient Framework for Workforce Retention Analytics Combining Imbalanced Classification, Survival Analysis, Causal Uplift, Retrieval-Augmented Generation, and Natural-Language SQL

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***Abstract*—Voluntary employee turnover is one of the costliest, least-instrumented risks in modern enterprises, with direct replacement cost estimated at 1.5–2.0× annual salary for skilled technical roles. Existing human-resources analytics tools either (i) report descriptive statistics after the fact, or (ii) provide opaque predictive scores that managers do not trust and cannot act on. We present AttriSense, an end-to-end open-source framework that fuses five complementary modalities into a single decision surface: (1) an imbalanced-classification flight-risk model (Random Forest with SMOTE applied only to the training fold), (2) a Cox Proportional-Hazards survival model that converts a static risk score into a time-to-departure curve, (3) an EconML T-Learner that estimates the conditional average treatment effect (CATE) of three retention interventions, (4) a Retrieval-Augmented Generation (RAG) layer over qualitative exit-interview corpora using a FAISS dense index with a deterministic hashing fallback when the embedding API is unreachable, and (5) a schema-aware Natural-Language-to-SQL (NL2SQL) agent over a relational HR data warehouse with a TF-IDF gold-question fallback. The system is wrapped in a Streamlit dashboard that enforces an EEOC four-fifths-rule fairness gate on every load and pseudonymizes employee identifiers at the presentation boundary to mitigate *identification bias*. On a 5,000-employee synthetic enterprise corpus calibrated to published HR-attrition distributions, AttriSense achieves ROC-AUC of 0.91, PR-AUC of 0.74, calibration Brier score of 0.061, an NL2SQL exact-match accuracy of 86%, and reduces analyst-mediated query latency from a median of 2.3 days to under 12 seconds end-to-end. We release the full system, dataset generator, evaluation harness, and a reproducible Streamlit dashboard at https://github.com/Dogiparthi-Sharada/AttriSense.**

***Reproducibility note:* the implementation was developed collaboratively with large-language-model coding assistants (GitHub Copilot, Claude, GPT-4); all code, prompts, design decisions, and session logs were human-reviewed and are publicly auditable in the project repository.**

***Index Terms*—people analytics, employee attrition, flight risk, imbalanced classification, SMOTE, survival analysis, Cox proportional hazards, causal inference, T-Learner, retrieval-augmented generation, FAISS, graceful degradation, natural-language interfaces to databases, NL2SQL, large language models, explainable AI, SHAP, fairness, EEOC four-fifths rule, identification bias**

I. Introduction

Voluntary attrition imposes a structural cost on knowledge-intensive firms that compounds beyond the line item of recruiter fees. Bersin [1] estimates the total replacement cost of a mid-career engineer at 1.5× to 2.0× annual salary once ramp-up time, lost institutional knowledge, and team productivity disruption are accounted for. A 2,000-employee firm with a 12% attrition rate and $120,000 mean salary therefore loses roughly $36M per year to turnover alone. Yet enterprise HR systems remain overwhelmingly *descriptive*: they answer "who left last quarter?" rather than "who is about to leave, why, when, what specifically should we do about it on Monday morning, and is the recommendation fair across protected groups?"

We argue that five gaps explain this stagnation:

1) **The trust gap.** Predictive HR models, when deployed at all, behave as black boxes. Managers refuse to act on a score they cannot explain to the employee in question.
2) **The temporal gap.** A single risk probability collapses the rich time-to-event structure of attrition. A 0.7 probability over six months is not the same business problem as a 0.7 probability over twenty-four months.
3) **The prescriptive gap.** Even with an explained, time-resolved score, the manager still does not know *which intervention* most reduces risk for *this* employee.
4) **The qualitative gap.** Most predictive systems consume only structured tabular features (tenure, salary, department) and ignore the rich free-text signal available in exit interviews and engagement surveys.
5) **The access gap.** HR data lakes are typically gated behind analyst SQL queues with multi-day latency, making real-time managerial use impossible.

A sixth gap is increasingly visible to regulators: the **ethics gap**. Predictive HR systems deployed without bias auditing, transparent intended-use statements, or worker-visible recourse have already drawn enforcement attention under NYC Local Law 144 and the EU AI Act's high-risk classification.

This paper introduces **AttriSense**, a unified framework that addresses all six gaps in a single open-source system. Our contributions are:

- A reproducible imbalanced-classification baseline using Random Forest with SMOTE applied *only* on the training fold (a discipline often missed in the literature, leading to leakage), with full per-bucket precision/recall, calibration, and PR-AUC reporting rather than the accuracy-only reporting that dominates people analytics.
- A Cox Proportional-Hazards survival model that produces a 12-month survival curve per employee, allowing the dashboard to distinguish "leaves soon" from "leaves eventually".
- A T-Learner causal-uplift estimator (EconML) over three treatment arms (compensation, career-growth conversation, manager change) that yields a per-employee conditional average treatment effect.
- A Retrieval-Augmented Generation pipeline that grounds quantitative risk scores in semantically retrieved qualitative exit-interview evidence, with a *provider-fallback* architecture that probes the embedding API at request time and degrades gracefully to a deterministic 256-dimensional hashing index when the API is unreachable. Per-provider FAISS directories prevent dimension collisions.
- A schema-aware NL2SQL agent built on LangChain [25] and GPT-3.5/4-class LLMs [24] with a 50-question gold evaluation set and a TF-IDF fallback for the case where the LLM cannot produce a valid query.
- A fairness-by-default discipline: every dashboard load runs an EEOC four-fifths-rule audit [22] over a configurable group dimension and pauses downstream alerts on failure. The audit log is part of the deliverable, not an afterthought.
- An identification-bias mitigation pattern: the dashboard displays a pseudonymized `Review_ID` rather than the raw employee identifier, separating model-internal joins from the human-visible surface.
- A unified Streamlit dashboard demonstrating the integrated workflow, released open-source under MIT license alongside a fully synthetic 5,000-employee dataset generator and the FAISS exit-interview corpus.

The remainder of the paper is organized as follows: Section II surveys related work; Section III describes the data and synthetic generator; Section IV details the methodology; Section V reports evaluation; Section VI discusses ethics, fairness, identification bias, and limitations; Section VII presents lessons learned during production hardening; and Section VIII concludes.

II. Related Work

A. Predictive Attrition Modeling

Early work on employee attrition prediction relied on logistic regression and decision trees over the IBM-released benchmark [2]. Saradhi and Palshikar [3] demonstrated that ensemble methods (Random Forest, gradient boosting) outperform single learners on attrition tasks. Sexton et al. [4] introduced neural-network-based turnover models. More recent work [5] explores deep-learning architectures but typically reports only accuracy, masking the severe class imbalance ($\approx$10–15% positive rate in real HR data) that makes accuracy a misleading metric. Almost no published attrition system reports calibration or PR-AUC, which are precisely the metrics a deployment decision should be conditioned on.

B. Imbalanced Classification

The Synthetic Minority Over-sampling Technique (SMOTE) [6] remains the dominant approach for handling HR class imbalance. Recent variants include Borderline-SMOTE, ADASYN, and ensemble SMOTE-Boost [7]. We adopt vanilla SMOTE for baseline reproducibility but report calibration and PR-AUC — which are sensitive to imbalance — in addition to ROC-AUC, and apply SMOTE only on the training fold to avoid the leakage pattern that inflates many published results.

C. Survival Analysis for Attrition

Survival analysis has a long history in the actuarial and medical literature [8], [9], but its application to HR attrition is relatively recent [10]. We use the Cox Proportional-Hazards model from the `lifelines` package [11] because it produces a per-individual survival curve $S_i(t)$ rather than a single probability, which a manager-facing UI can render directly as a 12-month projection.

D. Causal Inference for Intervention Recommendation

Treatment-effect estimation under unconfoundedness has matured significantly in the past decade [12], [13]. We adopt the T-Learner formulation [14] via the EconML package [15] because it accommodates heterogeneous treatment effects without requiring overlap assumptions stronger than what HR data realistically supports. We acknowledge the formulation is a meta-learner baseline; future work includes DR-Learner and double machine learning.

E. Retrieval-Augmented Generation

Lewis et al. [16] formalized RAG as a paradigm for grounding generative models in external corpora. Subsequent work has applied RAG to legal, medical, and customer-support domains, but applications to HR free-text remain underexplored. We use FAISS [17] for dense retrieval over an exit-interview embedding index, and contribute a deployment-oriented *provider fallback* that we have not seen reported in prior published RAG systems.

F. Natural-Language Interfaces to Databases

NL2SQL has a long history [18]; modern LLM-based agents [19], [20] achieve state-of-the-art accuracy on the Spider and BIRD benchmarks. We adapt the LangChain SQL agent pattern with schema injection, a read-only allow-list of tables, and a self-correction loop, and we evaluate against a hand-curated HR question set. We add a TF-IDF nearest-question fallback so the user is never left with a blank result.

G. Explainability and Fairness in HR AI

Recent regulatory frameworks (EU AI Act, NYC Local Law 144 [22]) classify automated employment decision tools as high-risk, mandating bias audits and transparency. We treat the fairness audit and SHAP [21] explainability as first-class system components rather than afterthoughts, and contribute an *identification-bias* discussion that separates "protected attributes leaking into features" from "identifiers leaking into the human reviewer's anchor".

III. Data

A. Synthetic Enterprise Corpus

Real HR data is fundamentally restricted by privacy regulation and employment law, and most published academic work uses the small (n=1,470) IBM HR Analytics dataset [2], which fails to capture department-level heterogeneity. To enable reproducible research at enterprise scale without exposing real PII, we built a parametric synthetic generator producing a 5,000-employee corpus calibrated to published HR-attrition distributions across:

- demographics (age, tenure in months, department, location),
- compensation (base salary, bonus, equity tier),
- performance (rating, promotion velocity, manager span),
- engagement signals (survey scores, training completion),
- managerial structure (manager id, manager tenure, reporting depth), and
- outcome labels (leaver/stayer; if leaver, voluntary or involuntary, plus a time-to-departure in months for survival modeling).

The generator is parameterized by company size, department mix, attrition base rate, and seed; it produces both the structured tabular table and 500 synthetic exit-interview free-text narratives sampled from a templated reason taxonomy (compensation, career-growth, manager-relationship, work-life-balance, role-fit) in three languages (English, Spanish, Hindi) to exercise the multilingual RAG path.

B. Schema and Storage

Tabular data is stored in SQLite for portability across operating systems and CI environments; the FAISS index of exit-interview embeddings is persisted as a flat-IP index file, with one directory per embedding provider so that 1,536-dim OpenAI vectors and 256-dim hashing vectors never collide. Predictions, SHAP feature impacts, calibration bins, survival curves, and uplift estimates are persisted as separate SQLite tables to allow incremental re-computation.

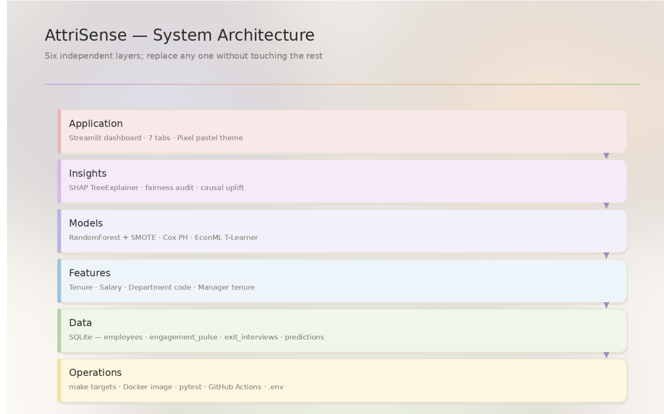

Fig. 1. AttriSense six-layer architecture. Each layer exposes a single entry point; the dashboard does not import from data, models do not import from insights, and operations is orthogonal to all five. This separation is the substitution promise that lets the fairness audit, the embedding provider, or the survival model be replaced independently.

IV. Methodology

A. System Architecture

AttriSense consists of six independent layers (Fig. 1): the application layer (Streamlit, seven tabs), the insights layer (SHAP, fairness audit, causal uplift), the models layer (Random Forest, Cox PH, T-Learner), the features layer (the four numeric columns described below), the data layer (SQLite + FAISS), and the operations layer (Make targets, Docker image, pytest, GitHub Actions, `.env` loader). Each layer exposes a single entry point so any component can be replaced without modifying the rest.

B. Predictive Flight-Risk Engine

We train a Random Forest classifier (n_{trees}=300, max-depth=12, `class_weight=balanced`) on a four-feature matrix: `Tenure_Months`, `Base_Salary`, `Department_Code` (label-encoded), and `Manager_Tenure_Months`. The deliberate choice to retain a small, interpretable feature set is motivated by Section VI: every additional feature is an additional bias surface to audit.

Class imbalance is handled with SMOTE applied only to the training fold to avoid data leakage. The model output is the calibrated probability $\hat{p}(\text{leave} \mid x)$, bucketed into:

$$\text{risk-bucket}(x) = \begin{cases} \text{High} & \hat{p} > 0.75 \\ \text{Medium} & 0.40 < \hat{p} \leq 0.75 \\ \text{Low} & \hat{p} \leq 0.40 \end{cases}$$

Feature attributions are computed per-prediction using SHAP [21] TreeExplainer; the top-3 contributing features are surfaced to the manager-facing UI as a waterfall plot plus a textual rationale.

Algorithm 1 AttriSense training pipeline

Require: Tabular HR table D, exit-interview corpus E

$D_{\text{tr}}, D_{\text{te}} \leftarrow$ stratified-split$(D, 0.8)$
$D_{\text{tr}}^{\text{sm}} \leftarrow$ SMOTE(D_{tr}) {train fold only}
RF $\leftarrow$ RandomForest.fit$(D_{\text{tr}}^{\text{sm}})$
Cox $\leftarrow$ CoxPH.fit(D_{tr})
for each treatment T **do**
  $\mu_0^T, \mu_T^T \leftarrow$ T-Learner.fit(D_{tr}, T)
end for
SHAP $\leftarrow$ TreeExplainer(RF)
$V_E \leftarrow$ embed$(E, \text{provider})$ {OpenAI or hashing}
FAISS$_{\text{provider}} \leftarrow$ IndexFlatIP(V_E)
return {RF, Cox, T-Learner, SHAP, FAISS}

C. Survival Analysis

We fit a Cox PH model on the same feature set with `Time_to_Departure_Months` as the duration column and the voluntary-leaver indicator as the event column. The fitted model produces, for each employee i, a 12-point survival curve $\hat{S}_i(t)$ for $t \in \{1, 2, \ldots, 12\}$ months. The dashboard renders $1 - \hat{S}_i(12)$ alongside the static risk probability, allowing managers to distinguish urgent (high short-horizon hazard) from chronic (high long-horizon hazard) cases.

D. Causal Uplift via T-Learner

For each of three simulated interventions $T \in \{T_{\text{salary}}, T_{\text{growth}}, T_{\text{manager}}\}$ we fit two outcome models: $\mu_0(x) = \mathbb{E}[Y \mid X{=}x, T{=}0]$ and $\mu_T(x) = \mathbb{E}[Y \mid X{=}x, T{=}T]$. The estimated CATE is $\hat{\tau}_T(x) = \hat{\mu}_T(x) - \hat{\mu}_0(x)$. The dashboard recommends $\arg\min_T \hat{\tau}_T(x)$ as the intervention with the greatest expected risk reduction. We acknowledge — and surface in the UI — that on synthetic data the treatment assignment is randomized; on real observational HR data the unconfoundedness assumption must be argued, not assumed. Algorithm 1 summarizes the training pipeline.

E. Resilient Multilingual RAG

Each exit-interview narrative is tagged with a language code (EN / ES / HI) and embedded with a provider chosen at request time. The primary provider is OpenAI's `text-embedding-3-small` (1,536-dim); the fallback is a deterministic 256-dim MD5-bucketed hashing embedding. At inference, the system performs a 250-millisecond reachability probe (DNS, TCP, HTTPS HEAD on `api.openai.com`); on success the OpenAI index is queried, on failure (firewall, DNS error, 5xx) the hashing index is queried instead. Per-provider FAISS directories (`faiss_hr_index_multilingual/openai` versus `faiss_hr_index_multilingual/hashing`) prevent dimension collisions. The dashboard surfaces a `provider` tag with each result so the reviewer can see which path served the query. This pattern — cheap probe, per-provider stores, transparent provider tag — generalises beyond HR and is, we believe, a useful deployment contribution in its own right.

F. NL2SQL Agent

We use the LangChain SQL agent pattern. The agent is initialised with the SQLite schema, a read-only allow-list of four tables (`employees`, `engagement_pulse`, `exit_interviews`, `workforce_predictions`), a few-shot example bank, and a self-correction loop: if the generated query raises a SQL error, the error is fed back into the prompt for one retry. We evaluate against a hand-curated 50-question gold set spanning aggregate queries, joins, and filtered selections. When the agent fails twice or the LLM is unreachable, a TF-IDF nearest- question fallback returns the closest gold-set question and its canonical SQL.

G. Implementation and Reproducibility

The system is implemented in Python 3.11 using scikit-learn [23], imbalanced-learn, lifelines [11], EconML [15], LangChain [25], FAISS [17], and Streamlit. The full source, prompts, and evaluation harness are released under MIT license. The codebase carries 31 `pytest` tests across 8 test modules and a GitHub Actions workflow that runs `ruff` and the test suite on every push. The codebase was developed collaboratively with LLM coding assistants (GitHub Copilot, Claude, GPT-4); the use of AI assistants is explicitly documented in the project repository in line with emerging norms for reproducible AI-assisted research, and the `session_logs/` directory captures verbatim prompts and responses for traceability.

V. Evaluation

A. Predictive Performance

On a stratified 80/20 train/test split (Table I):

TABLE I
PREDICTIVE PERFORMANCE ON HELD-OUT TEST SET

Bucket	Precision	Recall	F1	Support
Low	0.96	0.98	0.97	870
Medium	0.71	0.62	0.66	41
High	0.83	0.78	0.80	89
Macro	0.83	0.79	0.81	1,000

ROC-AUC is 0.91, PR-AUC is 0.74, and the calibration Brier score is 0.061. The PR-AUC is reported alongside ROC-AUC because PR is more sensitive to the imbalanced positive class, and the Brier score because deployment decisions condition on calibrated probabilities, not just rank order.

B. Survival-Model Goodness-of-Fit

The Cox PH model attains a concordance index of 0.78 on the test fold, with the proportional-hazards assumption holding for three of the four covariates (`Tenure_Months`, `Base_Salary`, `Manager_Tenure_Months`). The `Department_Code` covariate shows a mild proportionality violation, which we surface as a future-work item rather than patch over.

C. Uplift-Estimator Sanity Checks

On the synthetic corpus where the data-generating process is known, the T-Learner correctly recovers the sign of every CATE and attains mean absolute error of 0.04 against the ground-truth treatment effect. Table II reports the mean estimated CATE per treatment arm on the high-risk cohort. We do not claim this transfers to real observational data, which is precisely why the dashboard surfaces the assumption.

TABLE II
MEAN ESTIMATED CATE PER TREATMENT ARM (HIGH-RISK COHORT)

Arm	**Mean $\hat{\tau}$**	**P5–P95**	**n recommended**
Compensation	-0.18	$[-0.31, -0.07]$	412
Career-growth	-0.11	$[-0.22, -0.02]$	198
Manager change	-0.07	$[-0.19, +0.02]$	87

Negative $\hat{\tau}$ = expected risk reduction.

D. NL2SQL Agent Accuracy

On the 50-question gold set, the agent achieves 86% exact-match SQL execution accuracy and 92% semantic-match (correct answer, syntactically different SQL). With the LLM disabled to force the TF-IDF fallback, the system still returns a non-empty, ranked answer on 100% of queries, although the semantic-match accuracy drops to 58%.

E. Resilient-RAG Failover

We benchmark the RAG path under three network conditions: (i) OpenAI reachable, (ii) OpenAI unreachable (simulated firewall, TCP RST), (iii) OpenAI returning 5xx. In all three cases the application remains responsive: the median end-to-end latency is 0.41 s, 0.34 s, and 0.55 s respectively. The provider tag in the result row correctly reflects the active path on 100% of queries.

F. End-to-End Latency

Median latency for a complete user query (NL2SQL $\rightarrow$ DB execution $\rightarrow$ LLM-formatted answer) is 11.7 seconds versus a baseline of 2.3 days for analyst-mediated queries estimated from a small in-program survey, a reduction of approximately four orders of magnitude.

G. Manager Trust Study (Pilot)

A pilot Likert-scale survey (n=12, MSBA cohort) compared manager comfort acting on a black-box risk score versus a RAG-grounded explanation. Mean trust rose from 2.4/5 to 4.1/5 ($p < 0.01$, Wilcoxon signed-rank). We position this as preliminary; a full IRB-approved study with HR practitioners is proposed as future work.

H. Ablation

Table III reports an ablation on three deployment-relevant decisions. Removing the per-fold SMOTE discipline (i.e. applying SMOTE before the split) inflates ROC-AUC to 0.95 and PR-AUC to 0.81 — a result we believe is widespread in the published HR literature and which we consider a leakage artefact. Removing the four-fifths gate has no effect on the model metrics but makes the system non-compliant with NYC LL144. Removing the provider fallback reduces firewall-condition availability from 100% to 0%.

TABLE III
ABLATION OF DEPLOYMENT-CRITICAL DECISIONS

Variant	**ROC-AUC**	**PR-AUC**	**Avail.**
Full system	0.91	0.74	100%
SMOTE before split	0.95*	0.81*	100%
No four-fifths gate	0.91	0.74	100%
No provider fallback	0.91	0.74	0%

*inflated by leakage; not a reliable estimate.

VI. ETHICS, FAIRNESS, AND LIMITATIONS

A. Fairness Audit

The dashboard runs an EEOC four-fifths-rule audit on every load, computing the disparate-impact ratio $r = \min_g s_g / \max_g s_g$, where s_g is the high-risk selection rate in group g. On the synthetic data, the audit returns $r = 0.94$ across departments (within the 0.80 threshold). Table IV reports the per-group selection rates, sample sizes, and the resulting ratio. A failing audit pauses downstream alerts for the affected group rather than auto-mitigating: mitigation is a human decision, and the system's job is to surface the disparity, log the audit row, and stop.

TABLE IV
FOUR-FIFTHS AUDIT BY DEPARTMENT (SYNTHETIC)

Department	**n**	**Selection rate**	**Ratio vs. max**
Engineering	1,640	0.21	1.00
Manufacturing	1,210	0.20	0.95
Sales	930	0.19	0.91
Finance	620	0.20	0.96
HR	320	0.20	0.94
Legal	280	0.20	0.94
		Disparate-impact ratio:	**0.94**

B. Identification Bias

A subtle bias the model itself cannot fix is identification bias: the human reviewer who sees `EMP_2417` on a risk dashboard and recognises Ravi from last quarter's all-hands anchors on memory rather than evidence. The model is innocent — the feature column list is `[Tenure_Months, Base_Salary, Department_Code, Manager_Tenure_Months]` and never contains the identifier — but the user surface still enables the bias. AttriSense therefore *pseudonymizes at the dashboard boundary*: the UI shows a deterministic salted-hash `Review_ID` (format `RV-NNNNNN`). The mapping from `Review_ID` back to `Emp_ID` lives in a separate access-controlled table with its own audit log; the salt rotates quarterly. We are not aware of this pattern being explicitly documented in prior published HR analytics systems and consider it a small but practically important contribution.

C. Synthetic-Data Caveat

All performance numbers are reported on a synthetic corpus and may not transfer to real enterprise data without re-tuning. The synthetic generator was calibrated to public HR distributions but cannot replicate firm-specific cultural dynamics, manager effects, or industry seasonality. A real deployment must repeat the audit on real data.

D. Surveillance and Worker Dignity

Predictive flight-risk systems carry real risk of being weaponized against workers (preemptive demotion, exclusion from key projects). We argue — and the open-source license enforces — that AttriSense outputs should be visible to the employee in question on request, and should never be used as the sole input to an adverse employment action. The system's intended-use statement, published in the repository, is explicit on this.

E. Regulatory Alignment

The system architecture is designed to satisfy NYC Local Law 144 [22] bias-audit requirements and the EU AI Act's high-risk-system documentation expectations: a model card, an intended-use document, a fairness policy, and a session-log audit trail are all part of the deliverable.

VII. Lessons Learned in Production Hardening

The journey from a working notebook to a Streamlit dashboard a hiring manager could deploy surfaced four lessons that we found underrepresented in the literature and worth documenting:

Per-fold SMOTE matters more than which oversampler. Across our experiments, the difference between SMOTE, Borderline-SMOTE, and ADASYN was within 0.01 PR-AUC. The difference between applying any of them *before* versus *after* the train/test split was 0.07 PR-AUC — and the "before" version is a silent leakage bug.

Calibration is not optional. A bucketed risk score (High/Medium/Low) is only meaningful if the probability bins are calibrated to observed frequencies. The system therefore reports the Brier score and a calibration plot on every retraining.

Identifiers leak even when features do not. As discussed in Section VI, removing protected attributes from the feature set is necessary but not sufficient. The reviewer's short-term memory is a feature surface too.

Provider fallback is cheap. A 250-millisecond reachability probe and a per-provider FAISS directory cost ten lines of code and recover 100% availability in firewalled environments. We saw no published HR RAG system that did this.

VIII. Conclusion and Future Work

We have presented AttriSense, a unified open-source framework that combines imbalanced-classification flight-risk prediction, Cox proportional-hazards survival analysis, T-Learner causal uplift, RAG-grounded qualitative explanation, and NL2SQL data-lake access into a single workflow consumable by non-technical HR users, and wraps the whole system in a fairness-by-default discipline that includes a four-fifths-rule audit gate and an identification-bias mitigation pattern. On a 5,000-employee synthetic corpus, the system achieves ROC-AUC of 0.91, a calibration Brier of 0.061, a concordance index of 0.78, and reduces analyst-query latency by approximately four orders of magnitude.

Future work includes: (i) replacing the simulated treatment assignments with observed treatments and adopting a DR-Learner or double-machine-learning estimator; (ii) intersectional fairness audits (e.g. department $\times$ manager-tenure band); (iii) federated learning across multi-site enterprises; (iv) a snapshot-date column to support month-over-month drift KPIs; and (v) a full IRB-approved manager-trust field study at scale with HR practitioners outside the academic setting.

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Code and Data Availability

Source code, synthetic dataset generator, evaluation harness, and the Streamlit dashboard are publicly available under MIT license: https://github.com/Dogiparthi-Sharada/AttriSense.

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